

AN ORDER-14 COUNTEREXAMPLE TO CONJECTURE 4 OF PIOGE ET AL.

ABSTRACT. Pioge, Pietrasz, Seron, Novo, and Cerf disproved their Conjecture 4 with a positive semidefinite matrix of order 16 and reported no counterexample of order 15. We give an exact rank-two Gaussian-integer Gram matrix $A \in \mathcal{H}_{14}$ such that

$$\lambda_{\max}(\operatorname{Re} F_A) > \operatorname{per}(A), \quad (F_A)_{ij} = a_{ij} \operatorname{per}(A(i, j)).$$

Thus the order of the published construction decreases from 16 to 14; no claim is made for orders at most 13. The same witness gives an order-14 counterexample to Conjecture 3 of Pioge et al. It is also diagonally congruent to a rank-two orthogonal projection, yielding time-delay-induced anomalous bunching with 14 photons in 14 modes.

1. THE EXACT COUNTEREXAMPLE

Let $\mathcal{H}_n$ be the cone of $n \times n$ positive semidefinite Hermitian matrices. For $A = [a_{ij}] \in \mathcal{H}_n$, let $A(i, j)$ be obtained by deleting row i and column j , and define

$$(F_A)_{ij} = a_{ij} \operatorname{per}(A(i, j)).$$

Here $\operatorname{Re} F_A$ denotes the entrywise real part of F_A ; it is a real symmetric matrix. Conjecture 4 of Pioge et al. [1] asserts that

$$(1) \quad \lambda_{\max}(\operatorname{Re} F_A) \leq \operatorname{per}(A) \quad (A \in \mathcal{H}_n).$$

They disproved (1) in order 16 and reported an unsuccessful numerical search in order 15. The conjecture is therefore already known to be false; our contribution is the following order reduction.

Lemma 1. *Let $X = (x_{rj}) \in \mathbb{C}^{2 \times n}$. For $I \subseteq [n]$, let X_I be the submatrix formed by the columns indexed by I , and write*

$$\prod_{j \in I} (x_{1j}t + x_{2j}) = \sum_{k=0}^{|I|} q_{I,k} t^k.$$

If $|I| = |J| = m$, then

$$(2) \quad \operatorname{per}(X_I^* X_J) = \sum_{k=0}^m k!(m-k)! \overline{q_{I,k}} q_{J,k}.$$

2020 *Mathematics Subject Classification.* 15A15, 15A18.

Key words and phrases. permanent, positive semidefinite matrix, Gram matrix, counterexample.

Proof. Expand the two summands in each factor of the permanent. For fixed k and fixed k -subsets $S \subseteq I$ and $T \subseteq J$, exactly $k!(m-k)!$ bijections from I to J map S onto T . Summing first over such bijections and then over S and T gives (2). $\square$

Theorem 2. *Conjecture 4 of Pioge et al. fails in order 14, already for a rank-two Gaussian-integer Gram matrix.*

Proof. Let $A = X^*X$, where $X = X_R + iX_I \in \mathbb{Z}[i]^{2 \times 14}$ and

$$X_R = \begin{pmatrix} 1178 & -223 & 145 & 278 & -388 & -78 & -9 & 163 & -347 & 602 & -344 & 529 & -168 & 77 \\ -30 & -50 & 68 & -14 & 17 & 369 & 866 & -686 & -467 & -88 & -133 & 1199 & -794 & 247 \end{pmatrix},$$

$$X_I = \begin{pmatrix} 317 & -59 & 327 & 310 & 426 & -109 & 942 & 521 & -1308 & 672 & -52 & 117 & -497 & -368 \\ 838 & 744 & -92 & -307 & 438 & 270 & -403 & -274 & -107 & 907 & -728 & 386 & -41 & 166 \end{pmatrix}.$$

Then $A \in \mathcal{H}_{14}$, and

$$\det X_{\{1,2\}} = -350880 + 1045686i \neq 0,$$

so $\text{rank } A = 2$.

Take

$$v = (-24, 3, 24, 17, 14, -5, 13, -16, 22, -17, -12, -4, 8, -23)^T.$$

Then $\sum_i v_i = 0$ and $v^T v = 3622$. Put $p = \text{per}(A)$. Since no column of X is zero, the polynomial in Lemma 1 for $I = [14]$ is nonzero; hence (2) gives $p > 0$. Applying the lemma to A and to all cofactors $A(i, j)$ gives, in exact integer arithmetic,

$$\begin{aligned} \Delta &:= v^T (\text{Re } F_A) v - p v^T v \\ &= 188\,558\,958\,243\,265\,516\,176\,136\,227\,271\,677\,528\,258\,899\,031 \\ &\quad \cdot 10^{45} + 880\,644\,911\,529\,452\,618\,269\,391\,965\,199\,062\,697\,646\,489\,600 \\ &> 0. \end{aligned}$$

Therefore

$$\lambda_{\max}(\text{Re } F_A) \geq \frac{v^T (\text{Re } F_A) v}{v^T v} > p.$$

For reference, the certified Rayleigh ratio is

$$\frac{v^T (\text{Re } F_A) v}{p v^T v} = 1.001531003891338 \dots$$

$\square$

2. CONSEQUENCES

Corollary 3. *Conjecture 4 fails in every order $n \geq 14$.*

Proof. For $k \geq 0$, let $A_k = A \oplus I_k$. Block structure gives

$$\text{per}(A_k) = p, \quad F_{A_k} = F_A \oplus pI_k.$$

Padding v with k zeros preserves the strict Rayleigh inequality. $\square$

Corollary 4. *There is an entrywise positive correlation matrix $B \in \mathcal{H}_{14}$ such that*

$$\text{per}(A \circ B) > \text{per}(A).$$

Consequently, Conjecture 3 of Pioge et al. [1] also fails in order 14.

Proof. Let $\tau = v/\sqrt{3622}$ and, for $\varepsilon > 0$, define

$$(B_\varepsilon)_{ij} = \frac{1 + \varepsilon^2 \tau_i \tau_j}{\sqrt{(1 + \varepsilon^2 \tau_i^2)(1 + \varepsilon^2 \tau_j^2)}}.$$

This is the Gram matrix of the unit vectors

$$\frac{(1, \varepsilon \tau_i)^T}{\sqrt{1 + \varepsilon^2 \tau_i^2}},$$

so it is a correlation matrix and is entrywise positive for all sufficiently small $\varepsilon > 0$. Laplace expansion gives every row and column sum of F_A as p . Expanding at $\varepsilon = 0$ therefore yields

$$\text{per}(A \circ B_\varepsilon) = p + \varepsilon^2 (\tau^T (\text{Re } F_A) \tau - p) + O(\varepsilon^4).$$

The quadratic coefficient is positive by Theorem 2. $\square$

Lemma 5. *Let $D = \text{diag}(d_1, \dots, d_n)$ with $d_i > 0$, and set $C = DAD$. Then*

$$\text{per}(C) = \rho \text{per}(A), \quad F_C = \rho F_A, \quad \rho = \prod_{i=1}^n d_i^2.$$

Proof. The first identity follows by extracting d_i from row i and column i . For each i, j ,

$$\begin{aligned} (C)_{ij} \text{per}(C(i, j)) &= d_i d_j a_{ij} \left(\prod_{r \neq i} d_r \right) \left(\prod_{s \neq j} d_s \right) \text{per}(A(i, j)) \\ &= \rho (F_A)_{ij}. \end{aligned}$$

$\square$

Corollary 6. *There is a rank-two orthogonal projection $P \in \mathcal{H}_{14}$ violating (1). It yields time-delay-induced anomalous bunching with 14 photons in a 14-mode interferometer.*

Proof. Set

$$\begin{aligned} s &= 5\,975\,547\,235\,929, & a &= 91\,619\,936\,424\,609, \\ b &= 170\,572\,929\,732\,992, & c &= 119\,336\,587\,939\,769, \end{aligned}$$

and

$$W = \text{diag}(s, s, s, a, b, s, s, c, s, s, s, s, s, s).$$

Direct integer multiplication gives

$$XWX^* = \eta I_2, \quad \eta = 144\,241\,568\,073\,276\,041\,742.$$

Let $D = W^{1/2}$, $Y = \eta^{-1/2}XD$, and

$$P = Y^*Y = \eta^{-1}DAD.$$

Then $YY^* = I_2$, so P is a rank-two orthogonal projection. By Lemma 5, F_P and $\text{per}(P)$ are obtained from F_A and p by the same positive scalar factor. Hence the vector $\tau = v/\sqrt{3622}$ satisfies

$$\frac{\tau^T(\text{Re } F_P)\tau}{\text{per}(P)} = 1.001531003891338\dots > 1.$$

For $d \geq 0$, define the Gaussian time-delay Gram matrix

$$S_{ij}(d) = \exp(-d^2(\tau_i - \tau_j)^2).$$

Using the row and column sums of F_P and expanding at $d = 0$ gives

$$\begin{aligned} \frac{\text{per}(P \circ S(d))}{\text{per}(P)} &= 1 + 2 \left(\frac{\tau^T(\text{Re } F_P)\tau}{\text{per}(P)} - 1 \right) d^2 + O(d^4) \\ &= 1 + 0.003062007782676\dots d^2 + O(d^4). \end{aligned}$$

Thus the ratio exceeds 1 for all sufficiently small $d > 0$, which is temporal anomalous bunching in the criterion of Pioge, Novo, and Cerf [2].

Finally, the two rows of $Y \in \mathbb{C}^{2 \times 14}$ are orthonormal and can be completed to a unitary $U \in \mathbb{C}^{14 \times 14}$. For the first two output modes,

$$P_{ij} = \sum_{k=1}^2 \overline{U_{ki}} U_{kj}.$$

Injecting one photon into each input mode therefore realizes the claimed 14-photon, 14-mode configuration. This answers the fewer-than-16-photon question posed in [2]. $\square$

REFERENCES

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